



Introduction to Data Analytics

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Chapter 3: Probability Distributions

Binomial Probability Distribution

1. A probability that has only two outcomes.
2. Rules:
 - a) Must be a fixed number of Trials.
 - b) The trials must be Independent: The outcome of one trial does not affect the others.
 - c) Trial has only one two outcomes: **success or failure**.
 - d) The Probability of success remains the same in all trials

Binomial Probability Distribution

Notations:

- $n \rightarrow$ # of trials
- $p \rightarrow$ the prob. of a successful outcome in a single trial.
- $q \rightarrow$ the prob. of a failing.
- $X \rightarrow$ the number of successes that occur in the n trials
- $P(x) \rightarrow$ the prob. of getting “ x ” successes.

Binomial Probability Distribution

Practice:

You roll a fair six-sided die 5 times. Define a success as rolling a 4, and suppose we are interested in exactly 2 successes.

Identify each binomial parameter/notion for this situation and give its value/meaning:

- $n =$
- $p =$
- $q =$
- $X =$
- $P(X) =$

Binomial Probability Distribution

Practice:

You roll a fair six-sided die 5 times. Define a success as rolling a 4, and suppose we are interested in exactly 2 successes.

Identify each binomial parameter/notion for this situation and give its value/meaning:

- $n = 5$
- $p = \frac{1}{6}$,
- $q = \frac{5}{6}$,
- $X = 2$
- $P(X) = ??$

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Binomial Probability Distribution

Each of the five scenarios below satisfies the condition “exactly 2 successes (rolling a 6) in 5 rolls.”

- **Scenario 1:** (A) S × (B) S × (C) F × (D) F × (E) F
- **Scenario 2:** (A) S × (B) F × (C) S × (D) F × (E) F
- **Scenario 3:** (A) S × (B) F × (C) F × (D) S × (E) F
- **Scenario 4:** (A) F × (B) S × (C) S × (D) F × (E) F
- **Scenario 5:** (A) F × (B) F × (C) S × (D) S × (E) F

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Binomial Probability Distribution

The Binomial probability formula:

$$P(X = x) = \underbrace{\binom{n}{x}}_{\text{\#scenarios}} \times \underbrace{p^x}_{\text{probability of the } x \text{ successes}} \times \underbrace{q^{n-x}}_{\text{probability of the } n-x \text{ failures}}$$

with x = number of successes in n trials, p = success probability per trial, and $q = 1 - p$.

Where,

$$\binom{n}{k} = \frac{n!}{k! (n - k)!} = \frac{n(n - 1) \cdots (n - k + 1)}{k!}$$

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Binomial Probability Distribution

Numerical application: For our case $n = 5$, $p = \frac{1}{6}$, $q = \frac{5}{6}$, and $x = 2$; we now compute $P(X = 2)$ using the binomial formula:

$$\begin{aligned} P(X = 2) &= \binom{n}{x} p^x q^{n-x} \\ &= \binom{5}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3 \\ &= 10 \times \frac{1}{36} \times \frac{125}{216} \\ &= \frac{1250}{7776} \approx 0.1608 \text{ (about 16.08\%).} \end{aligned}$$

Binomial Probability Distribution

Practice:

Which of the following is false?

- a) There are n ways of getting 1 success in n trials, $\binom{n}{1} = n$.
- b) There is only 1 way of getting n successes in n trials, $\binom{n}{n} = 1$.
- c) There is only 1 way of getting n failures in n trials, $\binom{n}{0} = 1$.
- d) There are $n - 1$ ways of getting $n - 1$ successes in n trials, $\binom{n}{n-1} = n - 1$.

Binomial Probability Distribution

Practice:

Which of the following is false?

- a) There are n ways of getting 1 success in n trials, $\binom{n}{1} = n$.
- b) There is only 1 way of getting n successes in n trials, $\binom{n}{n} = 1$.
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- d) There are $n - 1$ ways of getting $n - 1$ successes in n trials, $\binom{n}{n-1} = n - 1$.

Binomial Probability Distribution

Practice:

Which of the following is not a condition that needs to be met for the binomial distribution to be applicable?

- a) The trials must be independent
- b) The number of trials, n , must be fixed
- c) Each trial outcome must be classified as a success or a failure
- d) The number of desired successes, k , must be greater than the number of trials
- e) The probability of success, p , must be the same for each trial

Binomial Probability Distribution

Practice:

Which of the following is not a condition that needs to be met for the binomial distribution to be applicable?

- a) The trials must be independent
- b) The number of trials, n , must be fixed
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Binomial Probability Distribution

Exercise 1:

Probability of rolling **at most two 4's**.

Binomial Probability Distribution

Exercise 1

If you flip a fair coin 5 times and roll a fair six-sided die 3 times,

- What is the probability of getting exactly 2 heads and rolling two 2's?
- Calculate the probability of obtaining at least 2 heads or at least one 6.
- Given that at least one 6 was rolled, what is the probability of obtaining exactly 3 heads?

Binomial Probability Distribution

Mean and standard deviation of binomial distribution

Binomial Probability Distribution

Mean and standard deviation of binomial distribution

Mean (μ): the expected number of successes from your procedure.

$$\mu = n \cdot p$$

Binomial Probability Distribution

Mean and standard deviation of binomial distribution

Mean (μ): the expected number of successes from your procedure.

$$\mu = n \times p$$

Proof:

Binomial Probability Distribution

Mean and standard deviation of binomial distribution

Standard deviation (μ): the typical spread of the number of successes around the mean;.

$$\sigma = \sqrt{n \times p(1 - p)}$$

Proof:

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Binomial Probability Distribution

Going back to the die example (success = “roll a 4”, $n = 5$):

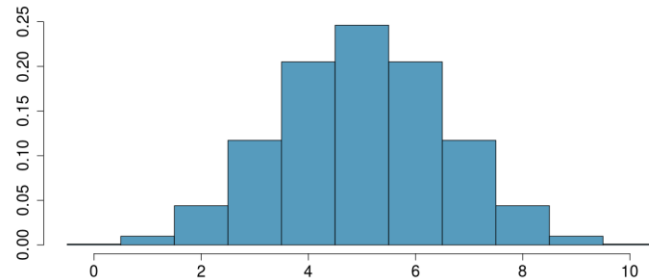
$$p = \frac{1}{6}, \mu = n \times p = \frac{5}{6} \approx 0.83, \sigma = \sqrt{n \times p(1 - p)} = \sqrt{5 \cdot \frac{1}{6} \cdot \frac{5}{6}} = \frac{5}{6} \approx 0.83.$$

So we'd expect **about 0.83 fours in 5 rolls**, with a **standard deviation ≈ 0.83** .

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Binomial Probability Distribution

Shapes of binomial distributions:

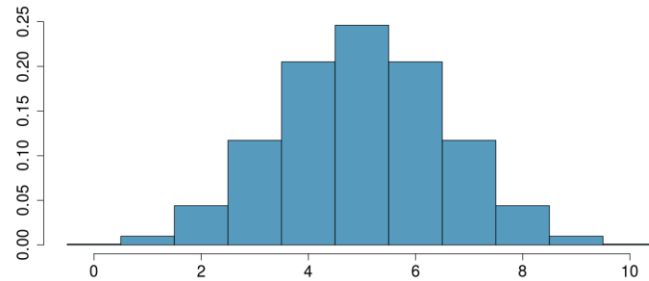


$n = 10$

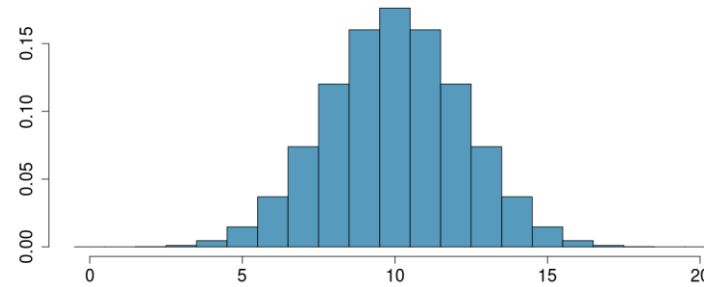
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Binomial Probability Distribution

Shapes of binomial distributions:



$n = 10$

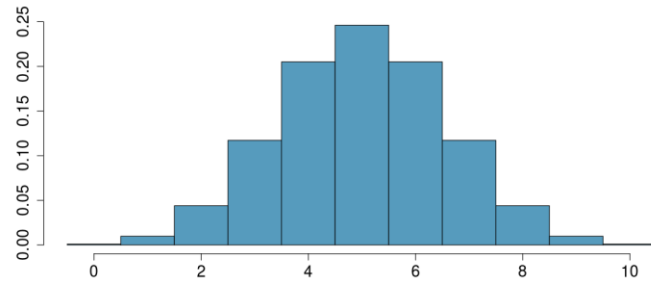


$n = 20$

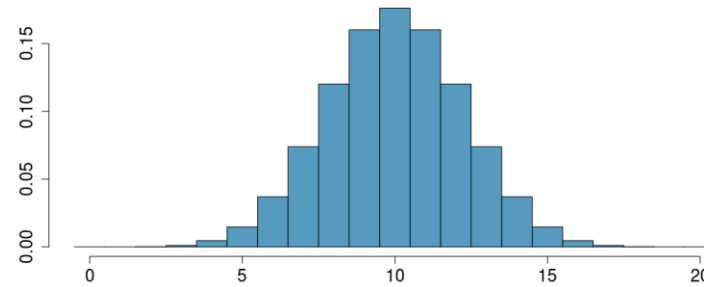
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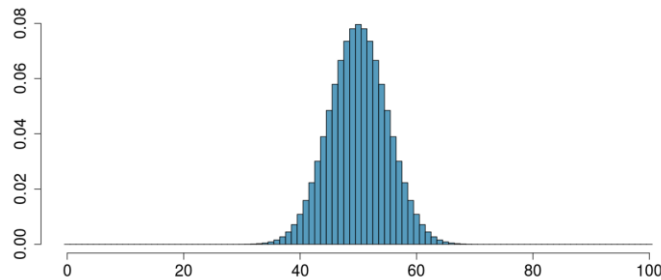
Shapes of binomial distributions:



$n = 10$



$n = 20$

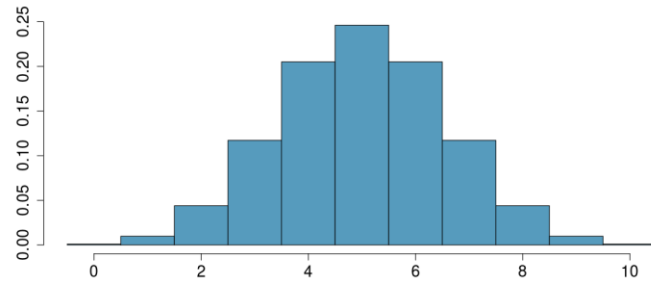


$n = 100$

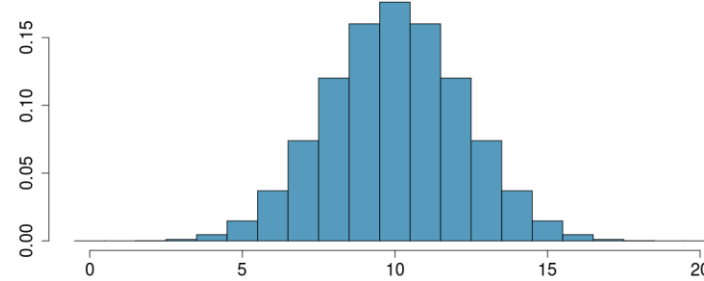
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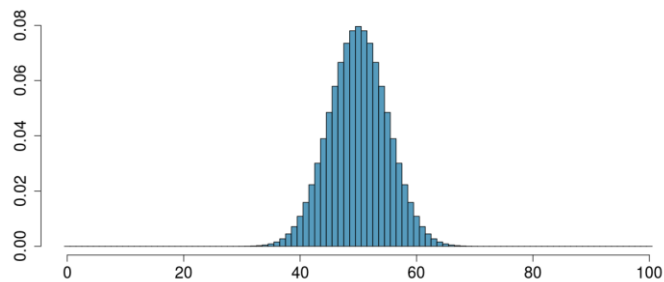
Shapes of binomial distributions:



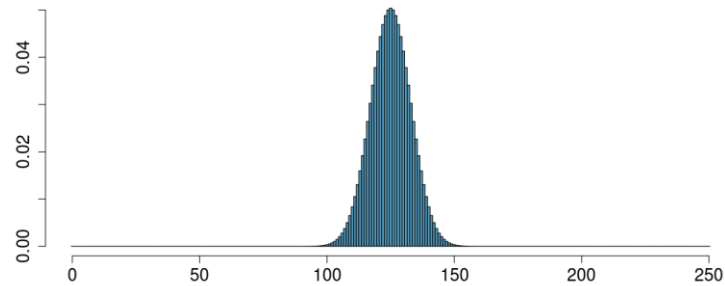
$n = 10$



$n = 20$



$n = 100$



$n = 250$

Binomial Probability Distribution

Low large is large enough?

The sample size is considered large enough if the expected number of successes and failures are both at least 10.

$$n \times p \geq 10 \text{ and } n(1 - p) \geq 10$$

Binomial Probability Distribution

Practice

Below are four pairs of Binomial distribution parameters. Which distribution can be approximated by the normal distribution?

- a. $n = 100; p = 0,95$
- b. $n = 25; p = 0,45$
- c. $n = 150; p = 0,05$
- d. $n = 500; p = 0,015$

Binomial Probability Distribution

An analysis of Facebook users

A recent study found that “Facebook users get more than they give”. For example:

- 40% of Facebook users in our sample made a friend request, but 63% received at least one request
- Users in our sample pressed the like button next to friends’ content an average of 14 times, but had their content “liked” an average of 20 times
- Users sent 9 personal messages, but received 12
- 12% of users tagged a friend in a photo, but 35% were themselves tagged in a photo

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Binomial Probability Distribution

This study also found that approximately 25% of Facebook users are considered power users. The same study found that the average Facebook user has 245 friends. What is the probability that the average Facebook user with 245 friends has 70 or more friends who would be considered power users? Note any assumptions you must make.

We are given that $n = 245$; $p = 0.25$, and we are asked for the probability $P(K \geq 70)$. To proceed, we need independence, which we'll assume but could check if we had access to more Facebook data.

$$\begin{aligned} P(X \geq 70) &= P(K = 70 \text{ or } K = 71 \text{ or } K = 72 \text{ or } \dots \text{ or } K = 245) \\ &= P(K = 70) + P(K = 71) + P(K = 72) + \dots + P(K = 245) \end{aligned}$$

This seems like an awful lot of work...

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Binomial Probability Distribution

When the sample size is large enough, a Binomial(n , p) distribution can be approximated by a Normal distribution with:

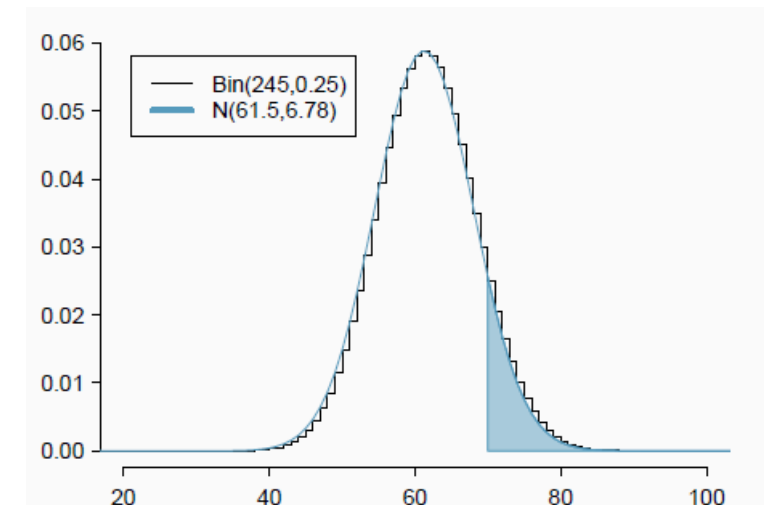
mean $\mu = n \times p$ and standard deviation $\sigma = \sqrt{p \times n(1 - p)}$

In the Facebook power users example: $n = 245$ and $p = 0.25$.

$$\mu = 245 \times 0.25 = 61.25$$

$$\sigma = \sqrt{245 \times 0.25 \times 0.75} \approx 6.78$$

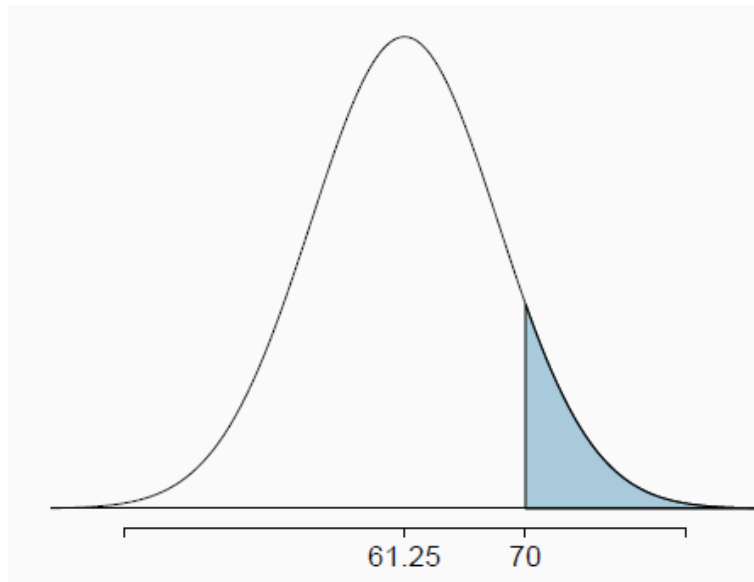
So: $\text{Bin}(n = 245, p = 0.25) \approx N(\mu = 61.25, \sigma = 6.78)$.



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Binomial Probability Distribution

What is the probability that the average Facebook user with 245 friends has 70 or more friends who would be considered power users?



$$Z = \frac{obs - mean}{SD} = \frac{70 - 61.25}{6.78} = 1.29$$

$$P(Z > 1.29) = 1 - 0.9015 = 0.0985$$

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Negative binomial distribution

- The negative binomial distribution gives the probability that the **k-th success** occurs on the **n-th trial**.
- It applies when:
 1. Trials are independent.
 2. Each trial is success/failure.
 3. The success probability **p** is the same on every trial.
 4. The last trial (the n-th) is a **success**.

(Note: the first three conditions are the same as for the binomial.)

Probability formula:

$$P(X = x) = \underbrace{\binom{n}{x}}_{\text{\#scenarios}} \times \underbrace{p^x}_{\text{probability of the } x \text{ successes}} \times \underbrace{q^{n-x}}_{\text{probability of the } n-x \text{ failures}}$$

where p is the probability that a single trial is a success.

All trials are assumed to be independent.